

Intraoperative Ultrasonography-Guided Positioning of Iodine 125 Plaque Brachytherapy in the Treatment of Choroidal Melanoma

Melinda Y. Chang, MD,¹ Mitchell Kamrava, MD,² D. Jeffrey Demanes, MD,² Min Leu, PhD,² Nzhde Agazaryan, PhD,² James Lamb, PhD,² Joel N. Moral, OD,¹ Robert Almanzor,¹ Tara A. McCannel, MD, PhD¹

Purpose: To report intraoperative ultrasonography-guided positioning of iodine 125 (I^{125}) plaques for brachytherapy of choroidal melanoma as a quality improvement measure.

Design: Retrospective, single-center, consecutive case-cohort study.

Participants: One hundred fifty consecutive patients with choroidal melanoma.

Methods: Patients with choroidal melanoma who were treated with I^{125} plaque brachytherapy from January 2007 through January 2011 with at least 6 months of clinical follow-up were included.

Main Outcome Measures: Patient and tumor characteristics at diagnosis were tabulated. The need for plaque repositioning if intraoperative ultrasonography showed the plaque to be either not centered on the tumor or if there was less than 1.0 mm of plaque margin beyond the tumor border was recorded. The rate of local treatment failure and occurrence of distant metastasis were determined.

Results: The average interval from surgery to last follow-up was 21.5 months. Fifty-four (36%) of 150 patients required plaque repositioning. Of tumors located in the macula, equator, and periphery, 15 (36.6%), 26 (36.6%), and 13 (34.2%) required repositioning. There was no case of local treatment failure during a mean follow-up of 21.5 months (range, 6–48 months). Clinical evidence of choroidal melanoma metastasis developed in 9 patients. The cumulative 2-year Kaplan-Meier rate of local treatment failure in the cohort was statistically lower compared with the Collaborative Ocular Melanoma Study, which did not require ultrasonography-guided plaque positioning.

Conclusions: Intraoperative ultrasonography identified the need to reposition I^{125} plaques to achieve centration and plaque margin (>1.0 mm) beyond the tumor border in 36% of eyes. Neither tumor size nor tumor location correlated with the need to reposition the plaque. There was no case of local treatment failure during follow-up in this series. Correct plaque position is an essential component of quality outcomes in brachytherapy. Intraoperative ultrasonography reduces geographic errors in placement in eye plaque therapy and may help to reduce local treatment failure in choroidal melanoma.

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Choroidal melanoma is the most common primary intra-ocular malignancy, with an incidence of 6 cases per million per year in the United States.¹ Fatal liver metastasis occurs in approximately 50% of cases. Iodine 125 (I^{125}) episcleral brachytherapy is the most common form of globe-conserving therapy for choroidal melanoma. The Collaborative Ocular Melanoma Study (COMS) demonstrated no significant difference in mortality between treatment with I^{125} brachytherapy versus enucleation after 12 years of posttreatment follow-up.²

However, published reports in the literature indicate a local treatment failure rate as high as 20% with I^{125} brachytherapy.^{3,4} The COMS reported a 5-year local recurrence rate of 10.3%.⁵ Local failure is associated with tumor and treatment morbidity consisting of either

the ocular side-effects of reirradiation or organ loss altogether by enucleation. Furthermore, higher rates of local treatment failure are associated with an increased risk of systemic metastasis and death.⁶

Modifying surgery to include intraoperative ultrasonography could significantly improve the accuracy of plaque positioning in relationship to the tumor, and thereby could improve the rate of local tumor control and quality of care. Others have reported on the use of intraoperative ultrasonography to aid in plaque positioning.^{7–12} This report describes the experience at the authors' center of I^{125} episcleral plaque repositioning after intraoperative ultrasonography-guided positioning of I^{125} plaque for brachytherapy of 150 patients with choroidal melanoma.